



St Catherine's
School
Waverley, Sydney

Student Number:

Year 12
Assessment Task 1

Mathematics Extension II

Time allowed: 55
minutes

General Instructions

- Attempt ALL questions
- Write your Student NUMBER at the top of this page and on the writing paper used

Question

Marks

Total marks

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Question 1.

If $z = 1 - \sqrt{3}i$ and $w = -1 + i$, find the Modulus and Argument of the following complex numbers

- (a) z (1m)
- (b) w (1m)
- (c) $\frac{z^5}{w^4}$ (2m)

Question 2

- (a) Solve for z : $z^5 = 1$, where z is a complex number (2m)
- (b) Show that if w is a complex root of $z^5 = 1$, then w^2, w^3 and w^4 are the other complex roots. (2m)
- (c) Factorise $z^5 - 1$ in the field of Complex Numbers (1m)
- (d) Factorise $z^5 - 1$ in the field of Real Numbers (2m)
- (e) Represent the five roots on an Argand diagram (1m)
- (f) Deduce that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$ (1m)

Question 3

Sketch the region on the Argand diagram containing all of the points representing the complex number z , such that

- (a) $-\frac{\pi}{4} \leq |z - i| \leq \frac{\pi}{4}$ (2m)
- (b) $z\bar{z} - 3(z + \bar{z}) \leq 0$ (2m)
- (c) $\text{Arg } z = \text{Arg}(z - 1)$ (2m)
- $\text{Arg } z = \text{Arg}(z - (1 + i))$

Question 4

Let α, β and γ be the roots of the equation $P(x) = 0$, where $P(x) = x^3 - 4x + 1$

Write down the cubic equation, whose roots are

- (a) α^2, β^2 and γ^2 (2m)
- (b) $\frac{1}{\alpha}, \frac{1}{\beta}$ and $\frac{1}{\gamma}$ (1m)

Also find the value of

- (c) $\alpha^3 + \beta^3 + \gamma^3$ (2m)

Question 5

If α is a root of multiplicity 3 of the polynomial equation $P(x) = 0$, (3m)

where $P(x) = x^4 + x^3 - 9x^2 + 11x - 4$, factorise $P(x)$

Question 6

- (a) Show that $1 + i$ is a root of $x^3 - 3x^2 + 4x - 2 = 0$ (2m)
- (b) Hence or otherwise find all the roots of the polynomial equation. (2m)

Question 7

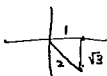
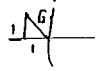
If the roots of $x^3 + kx^2 + lx + m = 0$ are in geometric progression, find the relationship between k, l and m . (3m)

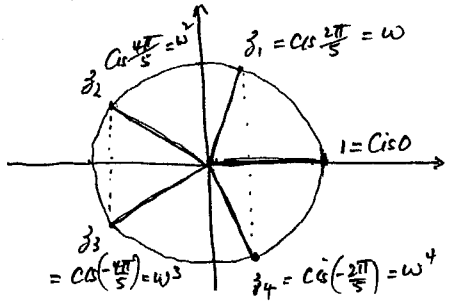
Question 8

One of the vertices of an equilateral triangle is the point represented by $1 + \sqrt{3}i$. (4m)

If the vertices of the equilateral triangle lie on a circle with centre the origin, find the other vertices and draw a neat sketch of the equilateral triangle clearly labelling the vertices.

END OF PAPER

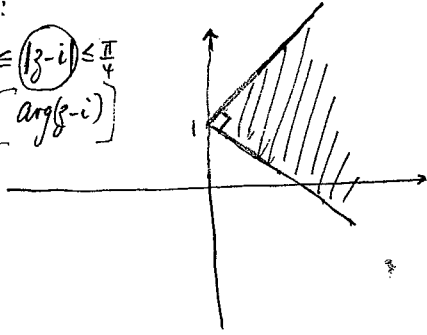
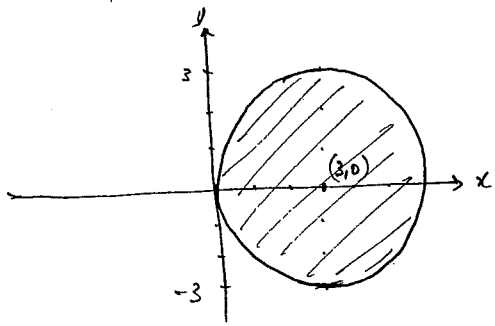
Solutions	Marks	Comments
<u>Question 1:</u> $z = 1 - \sqrt{3}i$ $w = -1 + i$ a) $z = 2$ $\text{Arg } z = -\frac{\pi}{3}$  b) $w = \sqrt{2}$ $\text{Arg } w = \frac{3\pi}{4}$  c) $z^5 = 32 \text{cis } -\frac{5\pi}{3}$ $w^4 = 4 \text{cis } 3\pi$ $\therefore \left \frac{z^5}{w^4} \right = \frac{ z^5 }{ w^4 } = \frac{32}{4} = 8$ $\text{Arg } \left(\frac{z^5}{w^4} \right) = \text{Arg } z^5 - \text{Arg } w^4$ $= -\frac{5\pi}{3} - 3\pi$ $= -\frac{14\pi}{3}$ $= -\frac{2\pi}{3}$ <u>Question 2:</u> a) $z^5 = 1$ let $z = r \text{cis } \theta$ $\therefore r^5 \text{cis } 5\theta = \text{cis } 0$ $\therefore r^5 = 1$ $r = 1$ $\cos 5\theta = 1$ $\sin 5\theta = 0$ $\therefore 5\theta = 2k\pi$ $k = 0, 1, 2, 3, 4$ $\therefore \theta = \frac{2k\pi}{5}$ $k = 0, 1, 2, 3, 4$ $\therefore$ solutions are $z = \text{cis } 0, \text{cis } \frac{2\pi}{5}, \text{cis } \frac{4\pi}{5}, \text{cis } \frac{6\pi}{5}, \text{cis } \frac{8\pi}{5}$ or $z = 1, \text{cis } \left(\pm \frac{2\pi}{5} \right), \text{cis } \left(\pm \frac{4\pi}{5} \right)$	① ① 1 ② 1 ②	

Solutions	Marks	Comments
b) let $w = \text{cis } \frac{2\pi}{5}$ then $w^2 = \text{cis } \frac{4\pi}{5}$ $w^3 = \text{cis } \frac{6\pi}{5}$ $w^4 = \text{cis } \frac{8\pi}{5}$ Which are the other complex roots c) $z^5 - 1 = (z-1)(z - \text{cis } \frac{2\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{6\pi}{5})(z - \text{cis } \frac{8\pi}{5})$ $= (z-1)(z - \text{cis } \frac{2\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{6\pi}{5})$ $= (z-1)\left[(z - \text{cis } \frac{2\pi}{5})(z - \text{cis } \frac{4\pi}{5}) \right]$ d) $z^5 - 1 = (z-1)(z - \text{cis } \frac{2\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{6\pi}{5})(z - \text{cis } \frac{8\pi}{5})$ $= (z-1)(z - \text{cis } \frac{2\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{4\pi}{5})(z - \text{cis } \frac{6\pi}{5})$ $= (z-1)(z^2 - 2\cos \frac{2\pi}{5}z + 1)(z^2 - 2\cos \frac{4\pi}{5}z + 1)$ e)  f) Sum of roots = 0 $\therefore 1 + \text{cis } \frac{2\pi}{5} + \text{cis } \frac{4\pi}{5} + \text{cis } \frac{6\pi}{5} + \text{cis } \frac{8\pi}{5} = 0$ $\therefore 1 + \text{cis } \frac{2\pi}{5} + \text{cis } \frac{4\pi}{5} + \overline{\text{cis } \frac{4\pi}{5}} + \overline{\text{cis } \frac{2\pi}{5}} = 0$ $\therefore 2\cos \frac{2\pi}{5} + 2\cos \frac{4\pi}{5} = -1 \therefore \cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$	② ① ② ① ①	

Course:

Marking Scheme for Task:

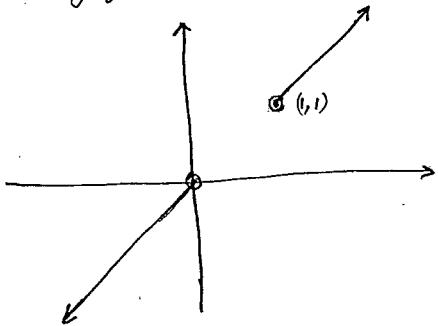
Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 3:</u> a) $-\frac{\pi}{4} \leq \arg(z-i) \leq \frac{\pi}{4}$ $\left[\arg(z-i) \right]$  b) $z\bar{z} - 3(z + \bar{z}) \leq 0$ $x^2 + y^2 - 3(2x) \leq 0$ $x^2 + y^2 - 6x \leq 0$ $x^2 - 6x + 9 + y^2 \leq 9$ $(x-3)^2 + y^2 \leq 9$ 	(2) 1 (2)	

Course:

Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 3 c)</u> $\arg z = \arg[z - (1+i)]$  <u>Question 4:</u> $P(x) = x^3 - 4x + 1$ a) let $y = x^2 \therefore x = \sqrt{y} = y^{\frac{1}{2}}$ $\therefore P(y) = (y^{\frac{1}{2}})^3 - 4(y^{\frac{1}{2}}) + 1 = 0$ $y^{\frac{3}{2}} - 4y^{\frac{1}{2}} + 1 = 0$ $y^{\frac{3}{2}} - 4y^{\frac{1}{2}} = -1$ $y^{\frac{1}{2}}(y - 4) = -1$ $y(y - 4)^2 = 1$ $y(y^2 - 8y + 16) = 1$ $y^3 - 8y^2 + 16y - 1 = 0$ b) let $y = \frac{1}{x} \therefore x = \frac{1}{y}$ $\therefore P(y) = (\frac{1}{y})^3 - 4(\frac{1}{y}) + 1 = 0$ $xy^3 - 1 - 4y^2 + y^3 = 0$ or $x^3 - 4x^2 + 1 = 0$	(2) 1 (2) 1	

Course:

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Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
(c) $\alpha^3 - 4\alpha + 1 = 0$ — (1) $\beta^3 - 4\beta + 1 = 0$ — (2) $\gamma^3 - 4\gamma + 1 = 0$ — (3) ①+②+③ $\alpha^3 + \beta^3 + \gamma^3 - 4(\alpha + \beta + \gamma) + 3 = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 = 4(\alpha + \beta + \gamma) - 3$ but $\alpha + \beta + \gamma = 0$ $\therefore \alpha^3 + \beta^3 + \gamma^3 = -3$. (2)	1	
<u>Question 5:</u> α root of multiplicity 3 $P(x) = x^4 + x^3 - 9x^2 + 11x - 4$ $P(1) = 0$ $P'(1) = 0$ $P''(1) = 0$ $\therefore x=1$ is the multiple root. $\therefore P(x) = (x-1)^3(x-\alpha)$ Since product of roots is -4 $\alpha = -4$ $\therefore P(x) = (x-1)^3(x+4)$ (3)	1	
<u>Question 6:</u> a) $(1+i)^3 = 1 + 3i - 3 - i$ $= -2 + 2i$ $(1+i)^2 = 1 + 2i - 1$ $= 2i$ $\therefore x^3 - 3x^2 + 4x - 2$ is $(1+i)^3 - 3(1+i)^2 + 4(1+i) - 2$ $= -2 + 2i - 6i + 4 + 4i - 2$ $= 0$ $\therefore (1+i)$ is a root. (2)	1	

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Course:

Marking Scheme for Task:

Academic Year: 2007-8

Solutions	Marks	Comments
<u>Question 6 b)</u> Since coefficients are real if $1+i$ is a root then $1-i$ is also a root. $\therefore (x-1-i)(x-1+i)$ is a factor $(x-1)^2 + 1$ is a factor $x^2 - 2x + 2$ is a factor. (1/2) $\therefore x^3 - 3x^2 + 4x - 2 = (x^2 - 2x + 2)(x-1)$ $\therefore$ roots are $1+i, 1-i, 1$ (2)	1	
<u>Question 7</u> let roots be $\frac{d}{c}, \alpha, \alpha c$ $\therefore$ product of roots $= \alpha^3 = -m$ — (1) Sum of roots $2\alpha = \frac{d}{c} + \alpha c + \alpha = l$ — (2) Sum of roots $= \frac{d}{c} + \alpha + \alpha c = -k$ — (3) now from (2) $\alpha(\frac{d}{c} + \alpha c + \alpha) = l$ $\therefore \alpha(-k) = l$ from (3) but $\alpha = -m^{1/3}$ from (1) $\therefore -m^{1/3} \cdot k = l$ cubing $mk^3 = l^3$ or $\underline{l^3 - mk^3 = 0}$ (3)	1/2	

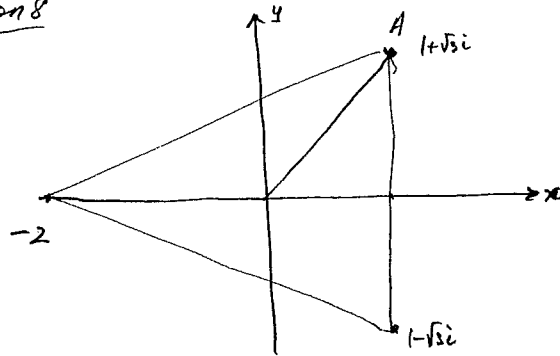
Course:

Marking Scheme for Task:

Solutions

Marks

Comments

Question 8

Angle subtended at centre by sides of an equilateral triangle is 120°

$$\begin{aligned} \therefore \text{next vertex} &= 1+\sqrt{3}i \times \text{cis } 120^\circ \\ &= (1+\sqrt{3}i)\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\ &= -\frac{1}{2} + \frac{\sqrt{3}}{2}i - \frac{\sqrt{3}}{2}i - \frac{3}{2} \\ &= -2 \end{aligned}$$

$$\begin{aligned} \text{next vertex} &= -2 \times \text{cis } 120^\circ \\ &= -2\left(-\frac{1}{2} + \frac{\sqrt{3}}{2}i\right) \\ &= 1 - \sqrt{3}i \end{aligned}$$

$\therefore$ vertices are $1+\sqrt{3}i$, $1-\sqrt{3}i$ and -2

1

2

3

4